

LAB # 01

Introduction to Linux Operating System

Objective

To study the Execution of various file/directory handling commands.

Lab Task:

1. Create multiple directories and subdirectories.

```
stewiedev@DESKTOP-RH97L4T: ~/directory
stewiedev@DESKTOP-RH97L4T:~$ mkdir directory
stewiedev@DESKTOP-RH97L4T:~$ ls
directory project01
stewiedev@DESKTOP-RH97L4T:~$ mkdir directory/subdirectory
stewiedev@DESKTOP-RH97L4T:~$ ls
directory project01
stewiedev@DESKTOP-RH97L4T:~$ cd directory
stewiedev@DESKTOP-RH97L4T:~/directory$ ls
subdirectory
stewiedev@DESKTOP-RH97L4T:~/directory$
```

2. Using man command(manual), explore atleast 5 extensions of each command i.e. **rmdir**, **mkdir**, **ls** and **pwd**. Implement them in any directory and insert screenshots.

```
clear(1)
NAME
  clear - clear the terminal screen
SYNOPSIS
  clear [-x] [-T terminal-type]
  clear -V
DESCRIPTION
  clear clears your terminal's screen and its scrollbar buffer, if any. clear retrieves the terminal type from the environment variable TERM, then consults the terminfo terminal capability database entry for that type to determine how to perform these actions.
  The capabilities to clear the screen and scrollbar buffer are named "clear" and "CU", respectively. The latter is a user-defined capability, applying an extension mechanism introduced in ncurses 5.0 (1999).
OPTIONS
  clear recognizes the following options.
  -T time produces instructions suitable for the terminal type. Normally, this option is unnecessary, because the terminal type is inferred from the environment variable TERM. If this option is specified, clear ignores the environment variables LINES and COLUMNS as well.
  -V reports the version of ncurses associated with this program and exits with a successful status.
  -x prevents clear from attempting to clear the scrollbar buffer.
PORTABILITY
  Neither IEEE Std 1083.1/The Open Group Base Specifications Issue 7 (POSIX.1-2008) nor X/Open Issues 7 documents clear.
  The latter documents tput, which could be used to replace this utility either via a shell script or by an alias (such as a symbolic link) to run tput as clear.
HISTORY
  A clear command using the terminfo database and library appeared in 2BSD (1979). Eighth Edition Unix (1985) later included it.
  The commercial Unix arm of AT&T adapted a different BSD program (tset) to make a new command, tput, and replaced the clear program with a shell script that called "tput clear".
  /usr/bin/tput $(t++TBI) clear 2/ /dev/null
  exit
  In 1989, when Keith Bostic revised the BSD tput command to make it similar to AT&T's tput, he added a clear shell script as well.
Manual page clear(1) line 1 (press h for help or q to quit)
```

```
rm(1)
NAME
  rm - remove files or directories
SYNOPSIS
  rm [OPTION]... [FILE]...
DESCRIPTION
  This manual page documents the GNU version of rm. rm removes each specified file. By default, it does not remove directories.
  If the -i or --interactive option is given, and there are more than three files or the -r, -d, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation. If the response is not affirmative, the entire command is aborted.
  Otherwise, if a file is unreadable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive option is given, rm prompts the user for whether to remove the file. If the response is not affirmative, the file is skipped.
OPTIONS
  Remove (unlink) the FILE(s).
  -f, --force
    ignore nonexistent files and arguments, never prompt
  -i prompt before every removal
  -I prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection against most mistakes
  --interactive=WHEN
    prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always
  --one-file-system
    when removing a hierarchy recursively, skip any directory that is on a file system different from that of the corresponding command line argument
  --no-preserve-root
    do not treat '/' specially
  --preserve-root=ALL
    do not remove '/' (default); with 'all', reject any command line argument on a separate device from its parent
  -r, -R, --recursive
    remove directories and their contents recursively
Manual page rm(1) line 1 (press h for help or q to quit)
```

```
ls(1)
NAME
  ls - list directory contents
SYNOPSIS
  ls [OPTION]... [FILE]...
DESCRIPTION
  List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cfGNU nor --sort is specified.
  Mandatory arguments to long options are mandatory for short options too.
  -a, --all
    do not ignore entries starting with .
  -A, --almost-all
    do not list implied . and ..
  --author
    with -l, print the author of each file
  -b, --escape
    print C-style escapes for nongraphic characters
  --block-size=SIZE
    with -l, scale sizes by SIZE when printing them; e.g., --block-size=M; see SIZE format below
  -B, --ignore-backups
    do not list implied entries ending with ~
  -c with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first
  -C list entries by columns
  --color=WHEN
    color the output WHEN; more info below
Manual page ls(1) line 1 (press h for help or q to quit)
```

```
mkdir(1)
NAME
  mkdir - make directories
SYNOPSIS
  mkdir [OPTION]... DIRECTORY...
DESCRIPTION
  Create the DIRECTORY(ies), if they do not already exist.
  Mandatory arguments to long options are mandatory for short options too.
  -m, --mode=MODE
    set file mode (as in chmod), not a-rwx - umask
  -p, --parents
    no error if existing, make parent directories as needed, with their file modes unaffected by any -m option.
  -v, --verbose
    print a message for each created directory
  -Z set SELinux security context of each created directory to the default type
  --context=CTX
    like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX
  --help display this help and exit
  --version
    output version information and exit
AUTHOR
  Written by David Mackenzie.
REPORTING BUGS
  GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
  Report any translation bugs to <https://translationproject.org/team/>
Manual page mkdir(1) line 1 (press h for help or q to quit)
```